

Homework 13

Please read Section 3.3 on Bayes'(s) formula. The formula itself is just before example 3.23.

1. Please work Exercise 55 (back-of-book answer 0.3343) from **Chapter 2**.

Seven students are chosen at random from a class with 17 boys and 13 girls. What is the probability that 4 boys and 3 girls are selected?

We think of this question as an urn problem. The probability is $C_{17,4} C_{13,3} / C_{30,7}$. That works out to be about 0.3343550447, which rounds to 0.3344. (I disagree with Durrett about the ten-thousandths place.)

2. Please work Exercise 56 (yuck!) from **Chapter 2**.

In a carton of 12 eggs, 2 are rotten. If we pick 4 eggs to make an omelet, what is the probability that we do not get a rotten egg?

We think of this question as an urn problem. The probability is $C_{10,4} / C_{12,4}$. That works out to be 14/33 or about 0.42.

I mention the common phrase "curate's egg" in class when talking about this problem. Here is the Wikipedia page where you can learn what that means:

https://en.wikipedia.org/wiki/Curate%27s_egg.

3. Please work Exercise 24 from **Chapter 3**. (This question is about commuting to work on public transportation.) Show and explain your work to make your problem-solving method clear.

John takes the bus with probability 0.3 and the subway with probability 0.7. He is late 40% of the time when he takes the bus, but only 20% of the time when he takes the subway. What is the probability that he is late for work?

We use the two-stage experiment (law of total probability) method.

Let A be the event that John is late for work. Let B be the event that he takes the bus, and let S be the event that he takes the subway. John takes one of these two public transportation routes to work (but not both), and so B and S form a partition of the sample space.

We are given $P(B) = 0.3$, $P(S) = 0.7$, $P(A|B) = 0.4$, and $P(A|S) = 0.2$. We thus have

$$P(A) = P(A|B)P(B) + P(A|S)P(S) = (0.4)(0.3) + (0.7)(0.2) = 0.12 + 0.14 = 0.26.$$

4. This exercise is what you were working on in class and matches one of Durrett's examples (but with slightly different numbers).

Breast cancer is relatively rare among women aged 40–50. About 1% of women in their 40s have breast cancer. Many authorities recommend breast-cancer screening with mammograms starting at age 40. For women in their 40s, mammograms detect about 87% of breast cancers. The “false positive” rate for mammograms given to women in their 40s is about 10%, meaning that about 10% of these women without breast cancer will receive a cancer diagnosis. (In the case of such a diagnosis, physicians follow up with more-invasive tests that have lower false-positive rates.)

Imagine a randomly chosen woman in her 40s. Let B be the event that the woman has breast cancer, and let A be the event that the woman is diagnosed with breast cancer by mammogram.

- a. What is the probability that the randomly chosen woman has breast cancer?

This is $P(B)$, which is 0.01, since 1% of women of this age have breast cancer.

- b. Imagine that the woman has a mammogram and is diagnosed with breast cancer on that basis. Express the probability that the woman has breast cancer given her diagnosis as a conditional probability using A and B . That is the probability that we would like to calculate.

$P(B|A)$ – that's the probability that the woman has breast cancer given that she has been diagnosed by mammogram.

- c. Use the Bayes's-formula method to calculate this probability. The probability is likely much smaller than you would have guessed. This phenomenon is what you should expect when testing for relatively rare conditions. The rarer the condition, the less information a positive test provides.

The general formula for conditional probability is $P(B|A) = P(B \cap A) / P(A)$. The Bayes' formula method is to do the following:

$$P(B \cap A) = P(A|B)P(B) \text{ (by the multiplication rule, which is formula 3.3 in Section 3.1)}$$

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c) \text{ (by the law of total probability)}$$

We are given that mammograms detect 87% of breast cancers for women of this age. That means $P(A|B) = 0.87$. ($P(A|B)$ is the probability of diagnosis of breast cancer with a mammogram given that a woman has breast cancer.)

We are given that mammograms have a false positive rate of 10% for women of this age. That means $P(A|B^c) = 0.10$. ($P(A|B^c)$ is the probability of diagnosis of breast cancer with a mammogram given that a woman does not have breast cancer.)

$P(B)$ is 0.01 and $P(B^c)$ is thus $1 - 0.01 = 0.99$.

Putting everything together, we find

$$P(B \cap A) = (0.87)(0.01) = 0.0087$$

$$P(A) = (0.87)(0.01) + (0.10)(0.99) = 0.0087 + 0.099 = 0.1077.$$

$$\text{Therefore, } P(B|A) = 0.0087 / 0.1077 \approx 0.0808.$$

Even with a breast cancer diagnosis by mammogram, the chance the woman has breast cancer is still small (about 8%). This fact leads to a lot of debate about which women should get mammograms. The tests themselves are somewhat expensive, and the follow-up for women who are incorrectly diagnosed can be painful. Of course, even if the great majority of diagnoses are false positives, mammography will catch some cancers early and improve the prognoses for treatment.

5. Two hunters shoot at a deer, which is hit by exactly one bullet. The first hunter hits targets with probability 0.3, and the second hits targets with probability 0.6. The event that the first hunter hits the deer and the event that the second hunter hits the deer are independent. Let's calculate the probability that the second hunter is the one who hit the deer.
- a. Let B be the event that the second hunter hits the deer. The probability we want is $P(B|A)$ for a certain event A . What is that event?

The event A is that exactly one hunter hits the deer.

- b. Use the Bayes's-formula method to calculate $P(B|A)$.

As above, we calculate $P(B|A)$ as $P(B \cap A)/P(A)$. We use the multiplication rule to calculate $P(B \cap A)$ and the law of total probability to calculate $P(A)$ as follows:

$$P(B \cap A) = P(A|B) P(B)$$

$$P(A) = P(A|B) P(B) + P(A|B^c) P(B^c)$$

We are given $P(B) = 0.6$, and so $P(B^c) = 0.4$.

To calculate $P(A|B)$, we think about the probability that exactly one hunter hits the deer given that hunter 2 hits the deer. That is the same as the probability that hunter 1 misses the deer given that hunter 2 hits the deer. We assume that the event that the first hunter hits

the deer is independent of the event that the second hunter hits the deer. Therefore, the probability that hunter 1 misses the deer given that hunter 2 hits the deer is simply the probability that hunter 1 misses the deer. That probability is 0.7.

To calculate $P(A | B^c)$, we think about the probability that exactly one hunter hits the deer given that hunter 2 misses the deer. That is the same as the probability that hunter 1 hits the deer given that hunter 2 misses the deer. We assume that the event that the first hunter hits the deer is independent of the event that the second hunter hits the deer. Therefore, the probability that hunter 1 hits the deer given that hunter 2 misses the deer is simply the probability that hunter 1 hits the deer. That probability is 0.3.

Putting everything together, we find $P(A | B)P(B) = (0.7)(0.6) = 0.42$ and $P(A) = P(A | B)P(B) + P(A | B^c)P(B^c) = 0.42 + (0.3)(0.4) = 0.54$.

Therefore, $P(B | A) = 0.42/0.54 = 42/54 = 21/27 = 7/9 \approx 0.78$.